

MODALEN III, Norway

WMO: 013260

Lat: 60.856N

Lon: 5.973E

Elev: 125

StdP: 99.83

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.6	-10.5	-15.5	1.0	-9.2	-14.0	1.1	-8.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	24.9	16.5	22.5	15.5	20.4	14.3	17.5	22.3	16.4	20.4	15.4	18.9	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	11.5	18.3	14.8	10.7	17.2	13.9	10.0	16.4	49.6	22.0	46.5	20.6	43.6	19.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-12.7	28.9	2.0	3.0	-14.1	31.0	-15.3	32.7	-16.4	34.4	-17.8	36.5		
(5) N/A	N/A	N/A	-13.8	18.4	2.4	1.2	-15.5	19.3	-16.9	20.0	-18.2	20.7	-19.9	21.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-1.1	-1.4	1.1	4.8	8.8	12.0	15.1	14.0	10.8	6.3	2.2	-0.7
	DBStd	6.70	4.05	3.93	3.02	2.89	3.21	2.70	3.21	2.45	2.58	3.37	3.59	4.58
	HDD10.0	1875	344	318	277	158	63	10	1	1	21	120	233	330
	HDD18.3	4496	603	551	535	406	297	189	111	135	226	372	483	588
	CDD10.0	435	0	0	0	2	25	71	158	126	46	7	1	0
	CDD18.3	13	0	0	0	0	1	1	10	2	0	0	0	0
	CDH23.3	146	0	0	0	0	5	18	111	12	0	0	0	0
	CDH26.7	27	0	0	0	0	1	1	25	0	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	2670	244	177	210	124	108	166	167	180	323	327	321	303
	PrecMax	3870	548	471	701	264	256	374	349	295	672	752	736	723
	PrecMin	1650	77	24	18	12	14	44	38	18	88	53	56	32
	PrecStd	634	137	121	168	57	70	87	98	84	150	166	174	197
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	7.7	9.1	17.0	23.0	25.9	29.8	25.1	19.9	16.9	11.5	9.6
		MCWB	5.3	4.7	5.0	10.1	13.7	15.5	18.3	17.0	14.1	12.4	8.8	7.6
	2%	DB	6.0	5.9	7.1	14.8	19.9	22.4	26.3	22.6	18.0	13.7	9.9	7.5
		MCWB	4.0	3.7	4.5	8.3	11.8	14.4	16.9	15.9	12.9	10.5	7.9	5.5
	5%	DB	4.6	4.6	6.0	12.2	17.6	20.0	23.7	20.7	16.5	12.2	8.5	6.1
		MCWB	3.0	2.9	3.9	7.0	10.7	12.6	16.3	15.2	12.9	10.3	6.7	4.5
	10%	DB	3.5	3.4	4.9	9.8	15.1	17.9	21.3	18.7	14.9	10.9	7.1	4.8
		MCWB	2.1	2.1	3.1	5.7	9.5	12.0	15.4	14.6	12.0	9.3	5.8	3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	5.8	6.5	10.0	14.7	16.6	18.9	18.7	16.2	13.3	10.0	8.0
		MCDB	7.1	7.0	7.9	14.6	22.3	22.8	26.5	21.7	17.3	14.2	10.8	8.6
	2%	WB	4.5	4.3	5.2	8.3	12.6	15.2	17.7	17.3	14.6	11.7	8.5	6.0
		MCDB	5.4	5.4	6.2	12.5	18.2	20.7	23.8	20.3	16.3	12.9	9.6	6.7
	5%	WB	3.4	3.2	4.4	7.2	11.3	14.0	16.8	16.1	13.6	10.8	7.2	4.7
		MCDB	4.3	4.2	5.4	10.3	16.1	18.1	21.7	18.8	15.3	11.9	8.0	5.6
	10%	WB	2.4	2.3	3.5	6.4	10.2	12.9	15.9	15.1	12.7	9.6	5.8	3.6
		MCDB	3.1	3.1	4.5	8.9	14.4	16.1	20.0	17.6	14.4	10.7	6.8	4.6
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.7	5.5	7.7	9.0	9.1	8.9	7.3	6.3	5.0	4.0	3.9
		MCDBR	4.7	5.1	6.4	11.9	13.9	14.1	14.2	11.0	9.1	7.0	5.8	5.2
		MCWBR	3.6	3.9	4.3	6.2	6.8	6.6	6.1	5.4	4.8	4.4	4.3	3.9
	5% WB	MCDBR	4.5	4.7	5.7	9.5	12.2	11.7	12.5	8.8	7.1	5.6	5.6	5.0
		MCWBR	3.7	3.9	4.4	5.5	6.3	6.0	5.7	4.9	4.7	4.2	4.7	4.2
Clear-Sky Solar Irradiance	taub	0.260	0.300	0.331	0.366	0.366	0.359	0.374	0.367	0.344	0.322	0.276	0.236	
	taud	2.082	2.183	2.183	2.178	2.348	2.420	2.407	2.421	2.476	2.414	2.184	1.999	
	Ebn at Noon	461	673	777	820	857	870	845	820	777	660	446	314	
	Edh at Noon	41	70	98	119	109	104	103	94	75	57	37	27	
All-Sky Solar Radiation	RadAvg	0.22	0.72	1.86	3.56	4.83	5.00	4.62	3.54	2.17	1.02	0.32	0.14	
	RadStd	0.04	0.17	0.33	0.49	0.59	0.59	0.48	0.35	0.28	0.15	0.05	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page